

Tissue Collection Protocol for Laser Capture Micro-dissection (LCM) of 13.5dpc Mouse Urethra and Hindgut

Cohn Group Protocols

Cohn & Barbazuk lab LCM protocol

Embedding

Preparation:

- Incubate isopentane at -80°C for ~ 1 hour.
- Incubate DEPC treated 1XPBS in the -20°C freezer for ~ 1 hour.
- Put dry ice in ice bucket, enough to just cover the bottom.
- Put dry ice in styro-foam box.
- Put ice in ice bucket.
- RNase-Zap all tools, petri dishes, molds and work area.
- Wear nitrile gloves.
- Add OCT (new container) to three small petri dishes. Keep on ice when not being used.
- Using a block of ice wrapped in black makes it easier to see the embryo.
- Pour cold isopentane into the dry ice bucket so that it covers the dry ice.
- Have tubes with lysis buffer (25mM NaOH) to extract the DNA for genotyping prepared.
- Have tubes with 1XPBS to collect extra tissue, upper half of the embryo, prepared.
- Have filter tip pipette box, any size, ready for use.
- Label molds and tubes.

Dissection:

- Place the embryos in ice-cold DEPC-treated 1XPBS in a petri dish sitting on a block of ice. Remove the embryos from the uterine-horn and their amniotic sacs.
- Place embryos in a fresh petri dish containing ice-cold DEPC-treated 1XPBS that is stored on a block of ice.
- Cut off the forelimb for PCR genotyping and the extra tissue – place in their respective tubes. The lower half of the embryo is collected for LCM.
- Keep the embryos on ice while they are not being dissected.
- Wash the embryos in OCT by placing it in the first OCT wash and swirl the embryo around using a pipette tip, 10 times. Transfer the embryo to the second OCT wash and repeat. Transfer the embryo to the third OCT wash and repeat. These washing steps remove excess water/PBS from the embryo to prevent the formation of ice crystals.
- Add OCT to the mold and place it on a chunk of dry ice to rapidly cool it down. Place the embryo in its correct orientation in to the OCT. Quickly flash freeze in the isopentane/dry ice bucket.

- Place the mold to the container with dry ice.
- Store at -80°C in a sealed Ziploc bag.

Sectioning

Preparation of the Cryostat:

- Turn cryostat to -20°C to -22°C for both the Object Temperature and Chamber Temperature.
 - If the cryostat has been used a lot that day it may be too warm and you either need to make it colder or wait until another day
- Clean the inside of the cryostat thoroughly with cotton gauze and 70% ethanol. Clean the outside of the cryostat with 70% ethanol, paying particular attention to the hand wheel and the glass window. Then clean with RNase Zap.
- Place the first mold to be cut into the cryostat and allow to equilibrate to temperature for ~30 minutes.
- RNase Zap Membrane Slides 1.0 PEN (Zeiss #415190-9041-000)
 - Dip the slides, one at a time, into a 50ml conical flask containing RNase Zap; then dip into a 50ml conical flask with DEPC-treated water and then in to a final 50 ml conical flask with DEPC-treated water. Allow to air dry on a foil-lined, RNase Zapped tray. The slides can be prepared the afternoon before so that they are dry by the time you use them.
- RNase Zap a new microtome blade the same way as the membrane slides.
- Spray the following tools with 70% ethanol and allow to air dry. Then spray with RNase Zap allowing them to dry before use.
 - Pencil for marking slides
 - Paint brushes
 - Razor blade
 - Glass anti-roll plate
 - Chuck
 - OCT bottle

Sectioning:

- RNase Zap gloves.
- Attach mold with tissue onto chuck. Fit blade into cryostat.
- Section at 10µm.
- Trim, discarding sections, until the tail disappears.
- Start collecting sections on to the membrane slides. In between picking up sections, keep slide at room temp. Move quickly! Collect 2 rows of 4 sections on each side. Once slide is full, allow sections to dry for ~1-2 minutes at room temperature. Store in slide box incubated on dry ice.
- Continue to collect the sectioned tissue until the genital tubercle disappears.
- Store slides at -80°C or proceed directly to staining.

Staining

Preparation:

- Day before staining: dissolve cresyl violet acetate powder (Acros Organics #229630050) at a concentration of 1% (w/v) in 50% ethanol & 50% RNase free water. Stir overnight at room temperature.
- Filter sterilize the cresyl violet acetate with a sterile Puradisc Syringe 25 Filter, 0.2µm, (Whatman #6780-2502) before first use.
- All ethanol solutions are made immediately prior to use with DEPC-treated RNase free water using an unopened bottle of 100% ethanol.

Staining:

- Stain the slides, 2 at a time (one after another), in sterile 50 ml conical flasks as follows:

95% ethanol for	30 seconds
75% ethanol for	30 seconds
50% ethanol (dip up & down) for	30 seconds
1% cresyl violet acetate in 50% ethanol	30 seconds
50% ethanol for	30 seconds
75% ethanol for	30 seconds
95% ethanol for	30 seconds
100% ethanol (dip up & down) for	30 seconds
100% ethanol	30 seconds
Xylene (dip up & down)	10X
Xylene for	5 minutes

Let slides air dry for 1-2 minutes and then place slide box on dry ice. Proceed directly to LCM. Carry out the stain & LCM procedures on same day.

Laser Capture Micro-dissection

- The PALM MicroBeam Laser microdissection system (Zeiss) is used for isolating high-purity tissue from the urethra & hindgut.
- RNA is purified from the sections using the RNeasy Micro Kit, (Qiagen #74004). RNase-DNase-free 0.65ml micro-centrifuge tubes (Corning #3208) are used during the collection & purification of the RNA.
- 50µl Buffer RLT with β-mercapto-ethanol (BME) is pipetted into the microfuge tube cap after it is loaded onto the tube collector. For each embryo one tube is prepared for the urethra and hindgut.
- Slides are kept on dry ice until needed. Each slide is removed from the dry ice and allowed to thaw on the edge of microscope stage for approximately 30 seconds. The slide is then positioned for viewing and micro-dissection.
- The urethra and hindgut are collected at 5X magnification. Once the urethra & hindgut are visualized together in the same section the components are micro-dissected, collected and stored in their respective microfuge tubes. Multiple urethra and hindgut dissections from the sections are collected into their sample tubes as quickly as possible.
- All sections before and after micro-dissection are photo documented.

- The number of sections collected per slide and the duration of collection, from start to finish, are recorded.
- The sections are micro-dissected sequentially using laser capture until all the sample tissue is collected.
- Caps are removed from the tube collector and labeled. Tubes are kept inverted.
- Vortex tubes upside down for 30 seconds. Store upside down at room temperature down until ready to proceed with RNA extraction. Do not store tubes on ice as the buffer will precipitate.

RNA extraction

- After collection, 300µl RLT buffer containing BME is added to each sample. Samples are vortexed for 1-2 minutes. The RNA is extracted according to the manufacturer's instructions.
- A DNase I digestion is carried out as follows.
 - Add 10µl DNase I stock solution to 70 µl Buffer RDD. Mix by inverting the tube. Add the DNase I incubation mix (80 µl) directly to the RNeasy MinElute spin column membrane. Place on the benchtop (20–30°C) for 15 minutes. Add 350 µl Buffer RW1 to the RNeasy MinElute spin column. Close the lid, and centrifuge for 15 s at $\geq 8000 \times g$. Discard the collection tube.
- RNA is eluted in 14µl RNase-free water.
- The RNA is analyzed for quality using the Agilent Bio-analyzer 2100.
- RNA integrity numbers above 7 are required for construction of the RNA-Seq libraries.

Solutions

1X PBS (1l)

100ml 10X PBS (Fisher Scientific #BP3994)

900ml ddH₂O

Autoclave at 121°C for 45 minutes.

DEPC treated 1X PBS (1l)

100ml 10X PBS (Fisher Scientific #BP3994)

900µl DEPC (Research Products International Corp #D43060-25.0)

Add ddH₂O to make 1l.

Let sit overnight.

Autoclave at 121°C for 45 minutes.

Consumables

Andwin Scientific Tissue-Tek™ CRYO-OCT Compound No.:4583 (Fisher Scientific #14-373-65)

RNaseZap® RNase Decontamination Solution (Ambion #AM9780)